

Artificial Intelligence - Deep Learning fundamentals

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Office hours in Google Meet or in person

<https://directorio.unizar.es/?#/tutoria?colectivo=PDI&codCentro=110>

M - 13.00-14.00h y de 1700 a 1900

J - 1600-18.00h

Please, isend me an email to book a slot 24h in advance to confirm (I am part time at Unizae,my week schedule changes, some weeks I may need to re-locate office hours) and specify if it's in person or online.

In this part

- Fundamentals of DL
 - Introduction & Shallow netes
 - Deep nets
- Convolutional NN
- Unsupervised
- Generative models. GANs
- Transformers
- Recent DL challenges. Foundation models.

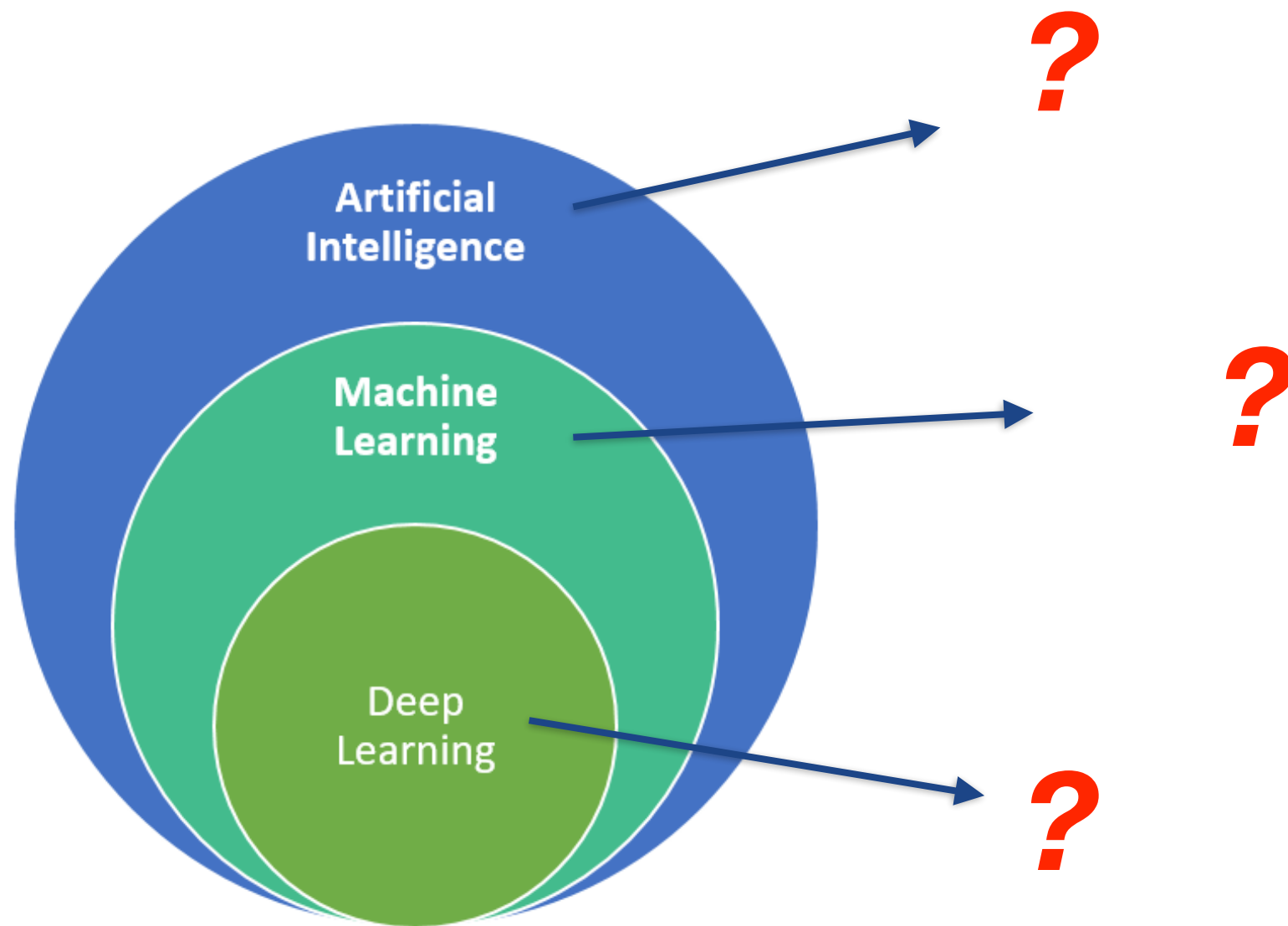
Let's know each other a little bit better

- Myself
- You
 - What is your background
 - What do you expect from this course
- Little survey about the fundamentals

Today's class

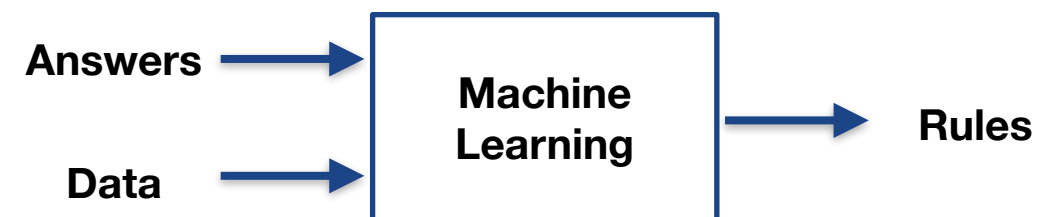
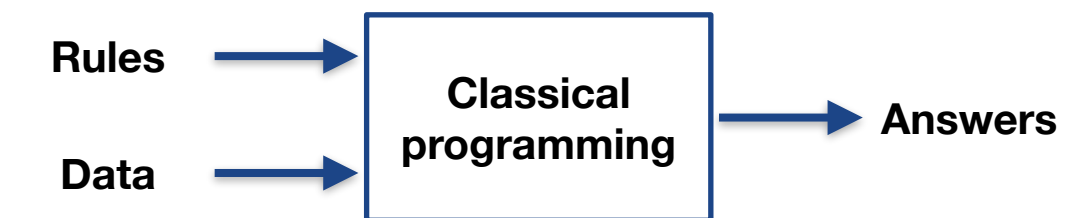
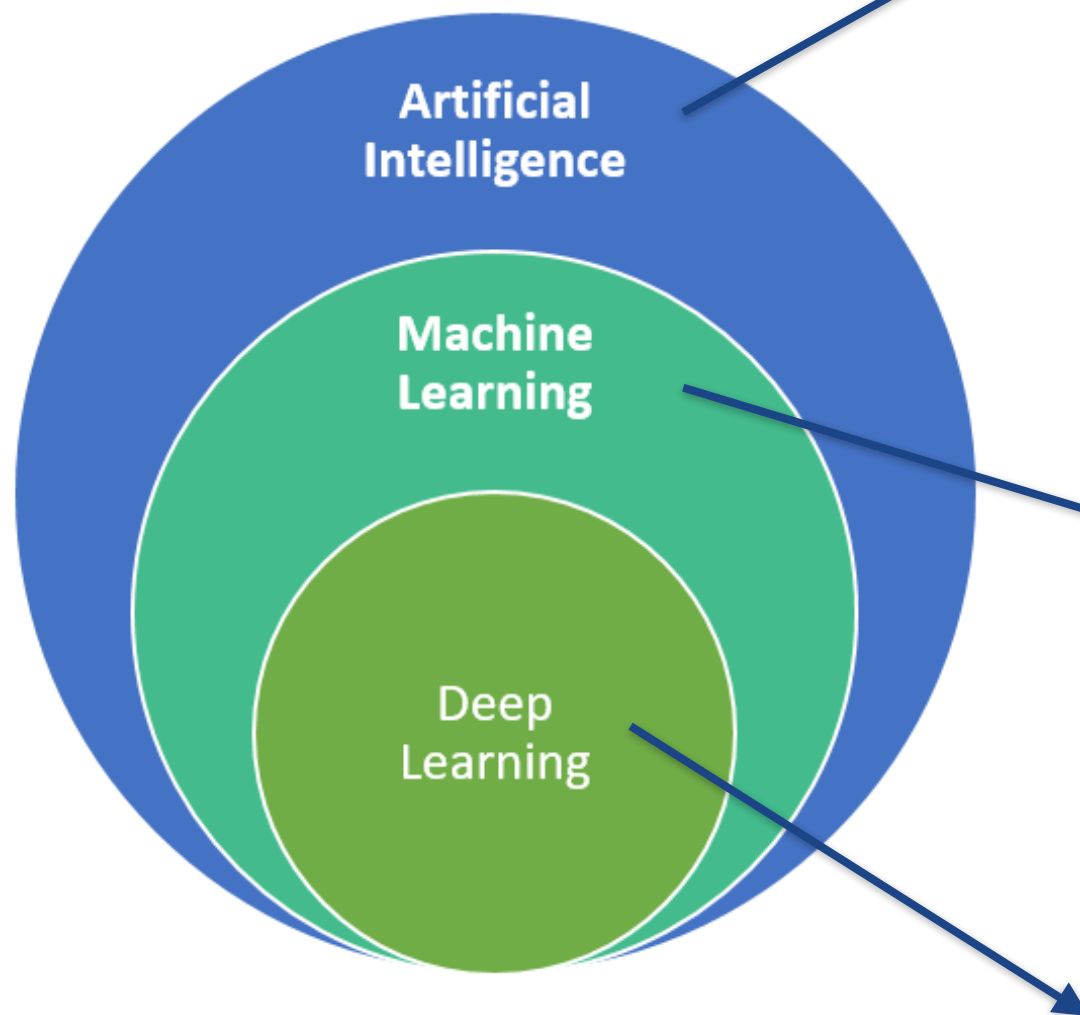
- **What's deep learning?**
 - **Related topics**
 - **DL pipeline**
- Fundamentals of DL
 - Review basic concepts
 - NN and DNN

What's *deep learning*? (and what's not ...?)



What's *deep learning*? (and what's not ...?)

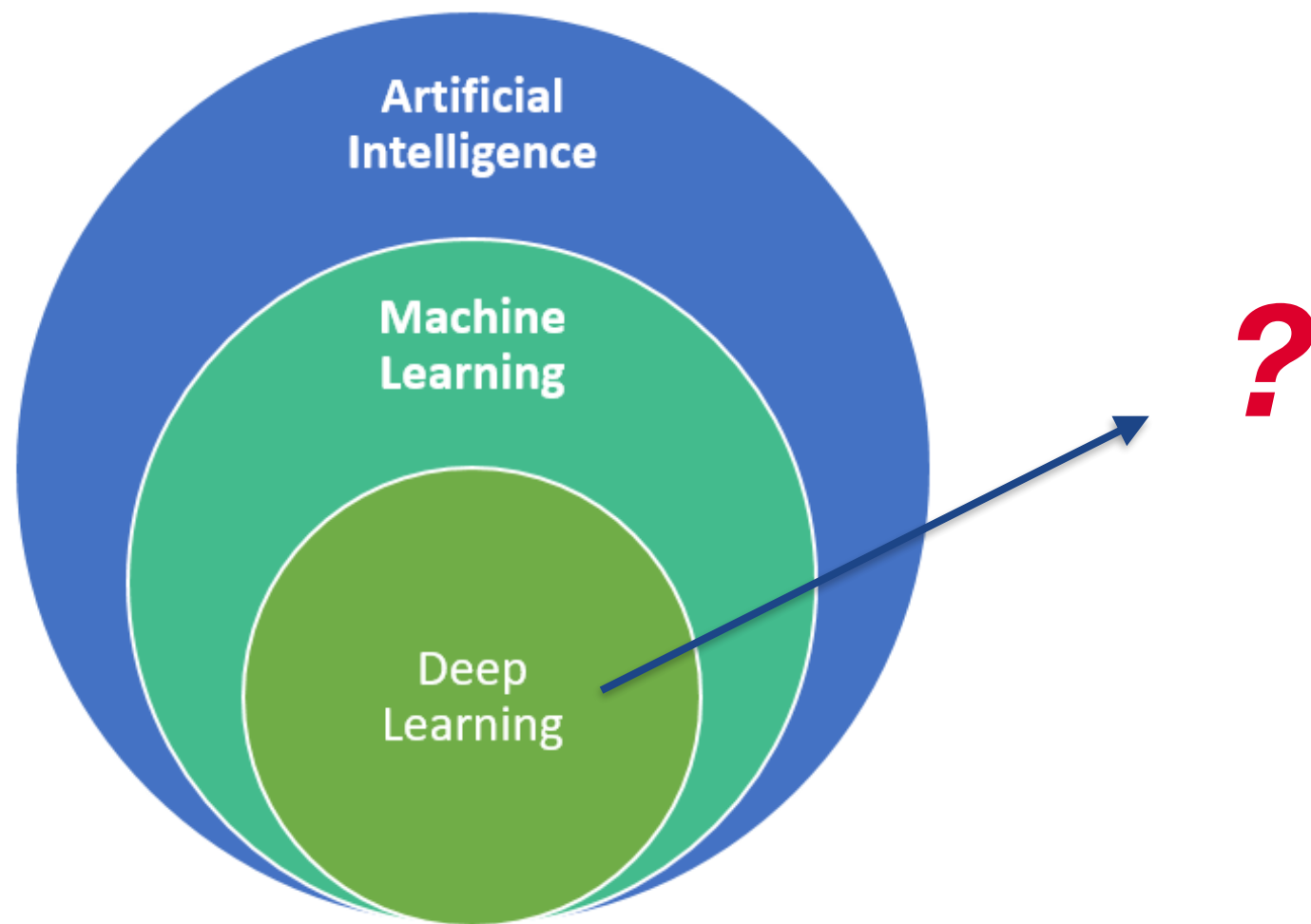
automate tasks that *usually* people performs
(any non-machine learning example?)



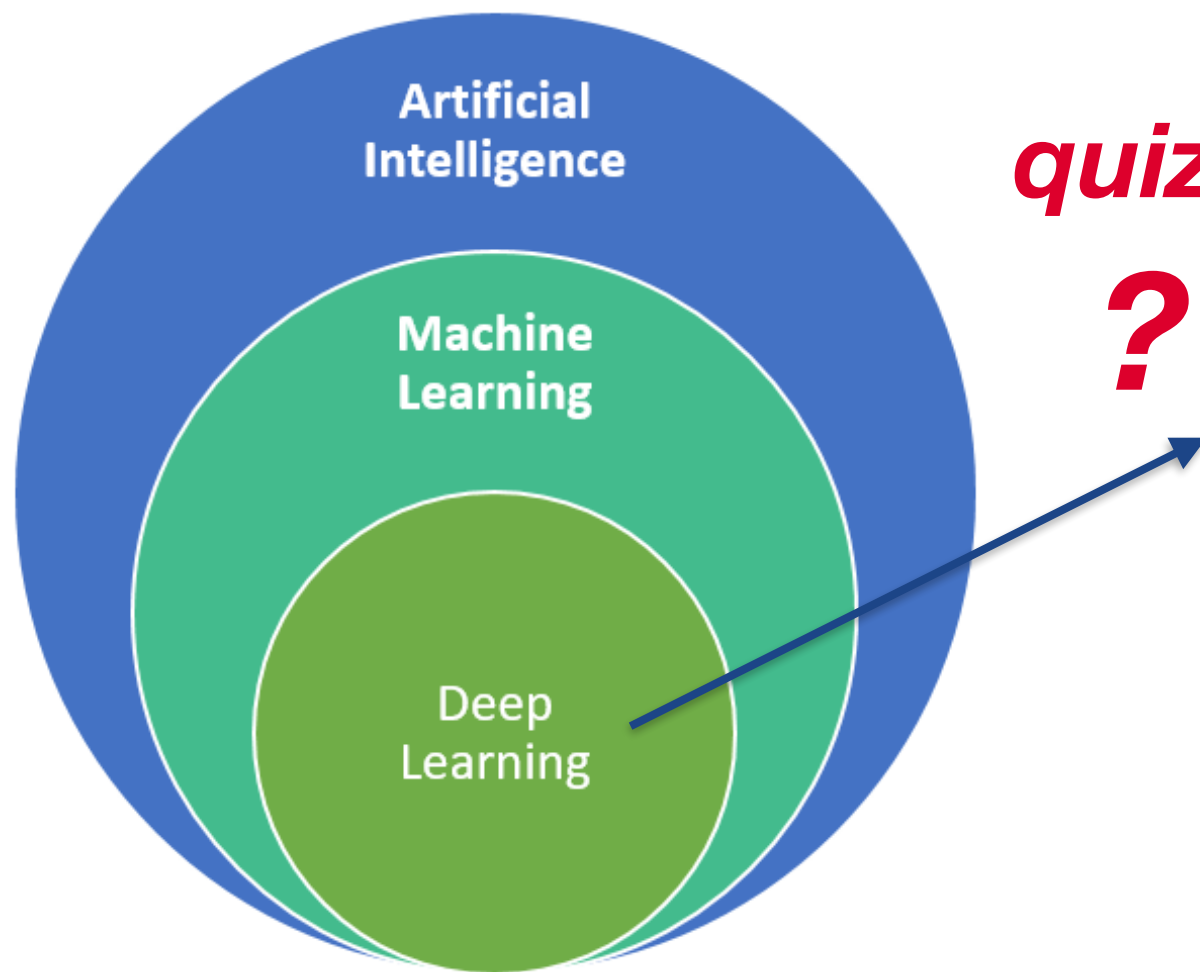
when that is achieved
by learning from data (examples)

using certain type of techniques
(*with a lot of layers*)

What's *deep learning*? (and what's not ...?)



What's *deep learning*? (and what's not ...?)

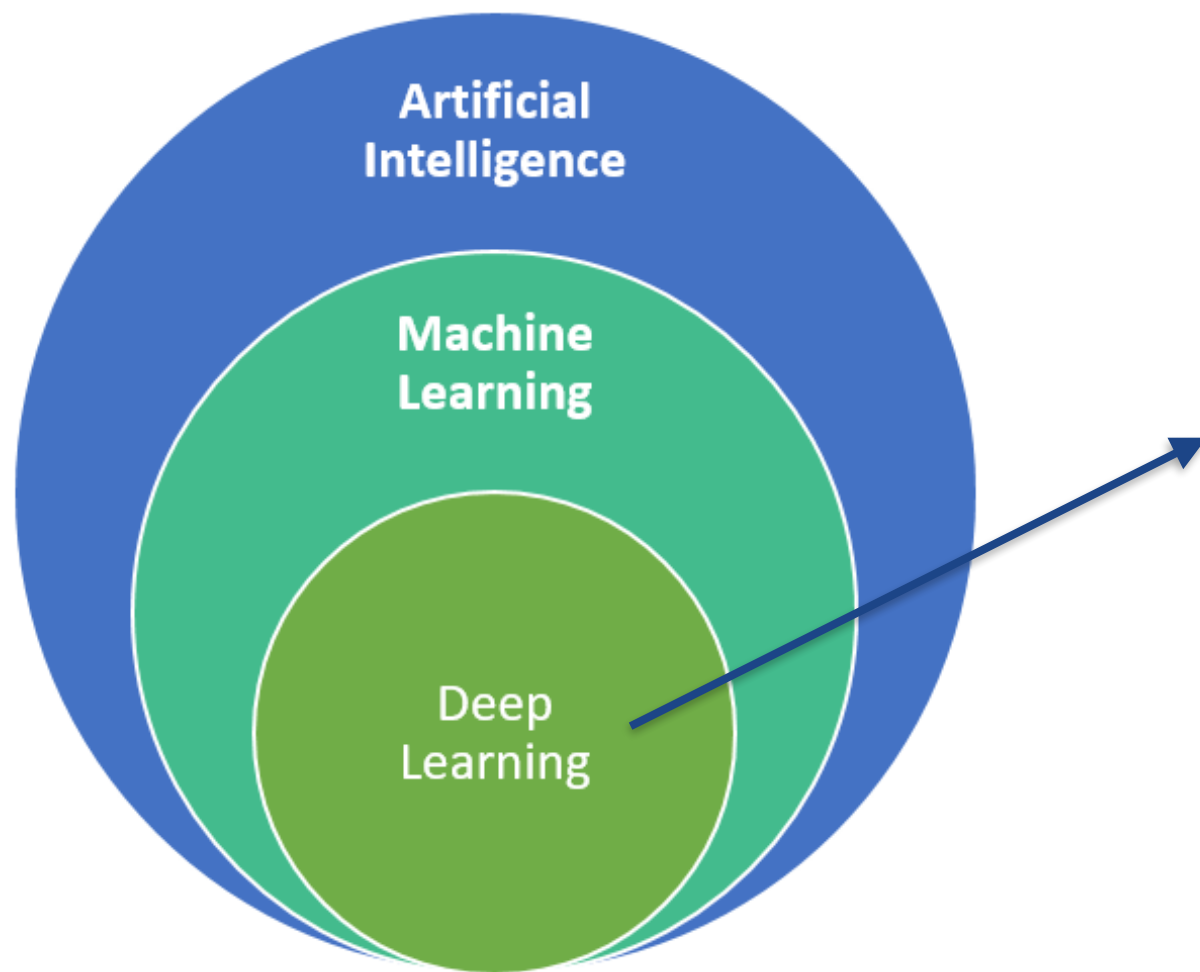


quizz

?

- Model of our brain
- Models with numerous layers (depth of N layers)
- Deep understanding of the problem/solution
- Learning *useful* representations from data (hierarchical representations of the data)

What's *deep learning*? (and what's not ...?)

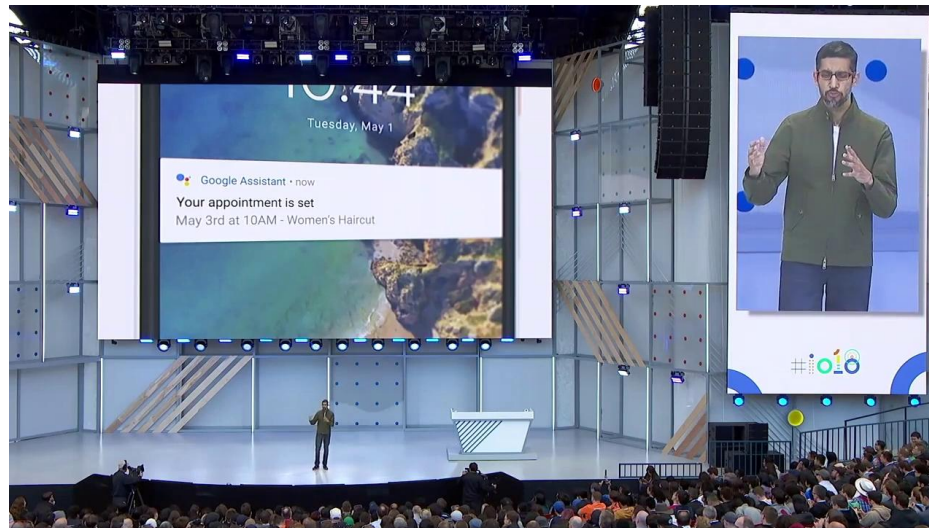
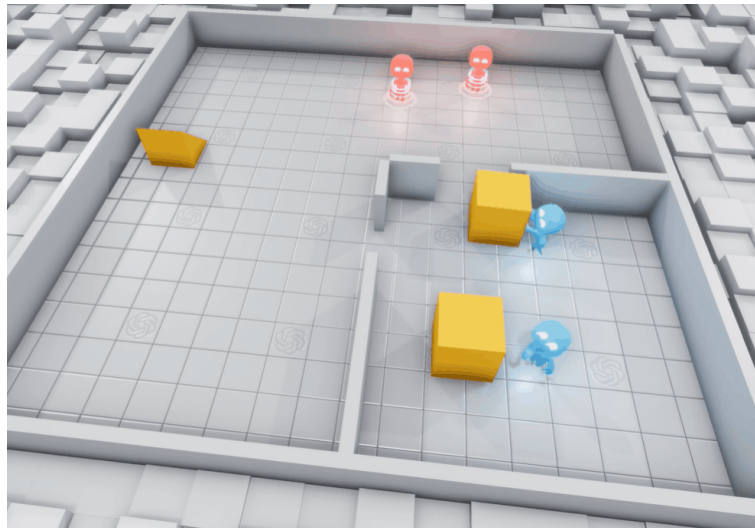


- ~~Model of our brain~~
- Models with numerous layers (depth of N layers)
- ~~Deep understanding of the problem/solution~~
- Learning *useful* representations from data (hierarchical representations of the data)

What can we do with deep learning?

- Some examples ... (and a bit of “recent” history)

What can we do with deep learning?



And what not ... :-)



Some examples of successful DL applications ...

- **Voice** recognition and Natural **Language** Processing



Hi, how can I help?

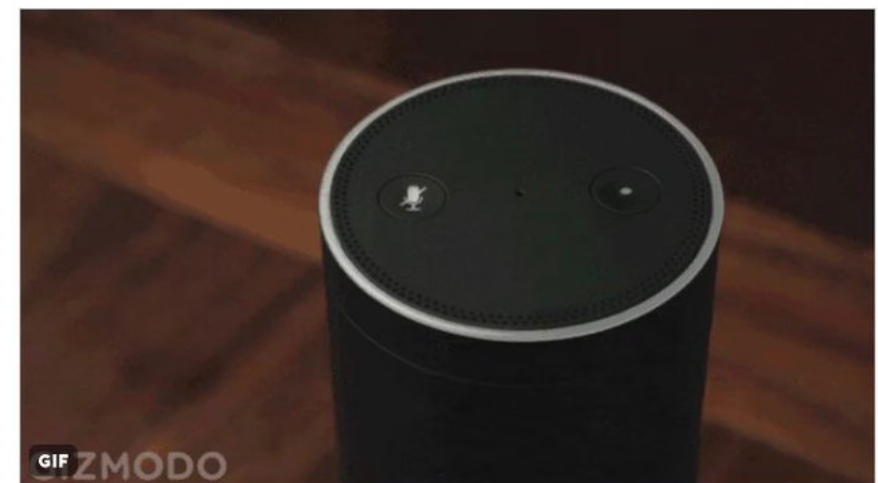


Not only visual tasks ...

Jeff Bezos: "Alexa, buy me something from Whole Foods."

Alexa: "Sure, Jeff. Buying Whole Foods now."

Jeff Bezos: "WHA- ahh go ahead."



6/16/17, 9:23 AM

Tweet your reply



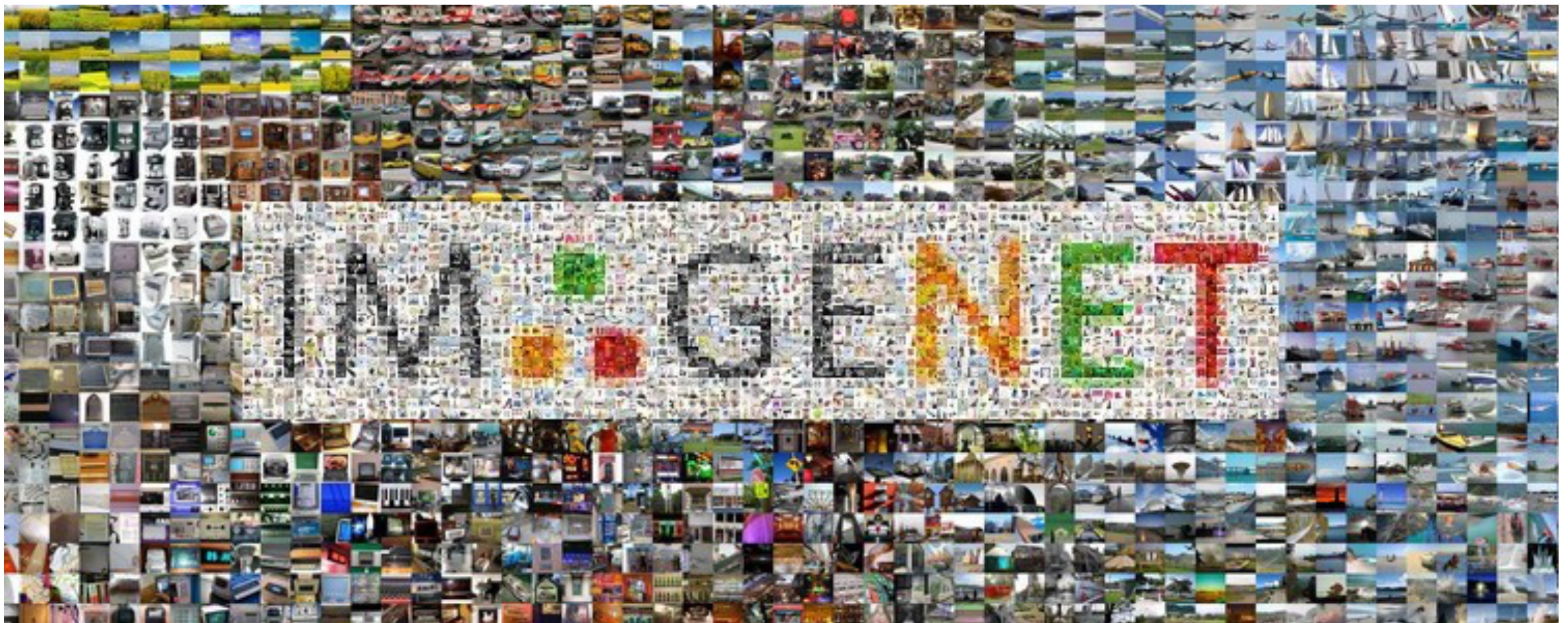
Some examples of successful DL applications ...

- **Object recognition**

Objects

- AlexNet. Competed in the ImageNet Large Scale Visual Recognition Challenge in 2012
- ImageNet: over 10 million annotated images

<http://image-net.org>

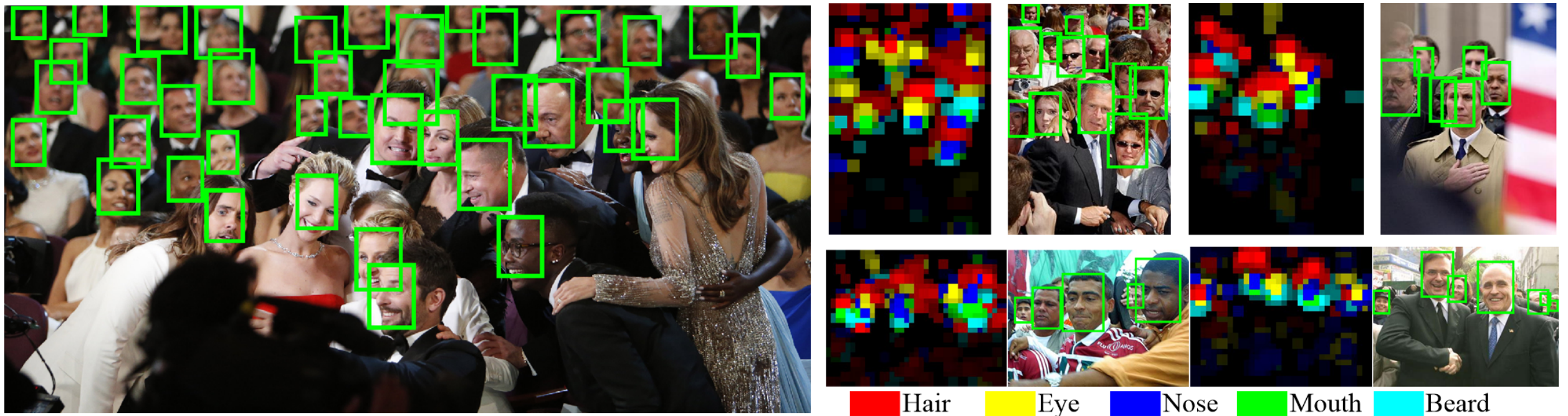


Some examples of successful DL applications ...

- **Face detection**

People

- CelebFaces Attributes Dataset (CelebA). Large-scale face attributes dataset, more than 200K celebrity images



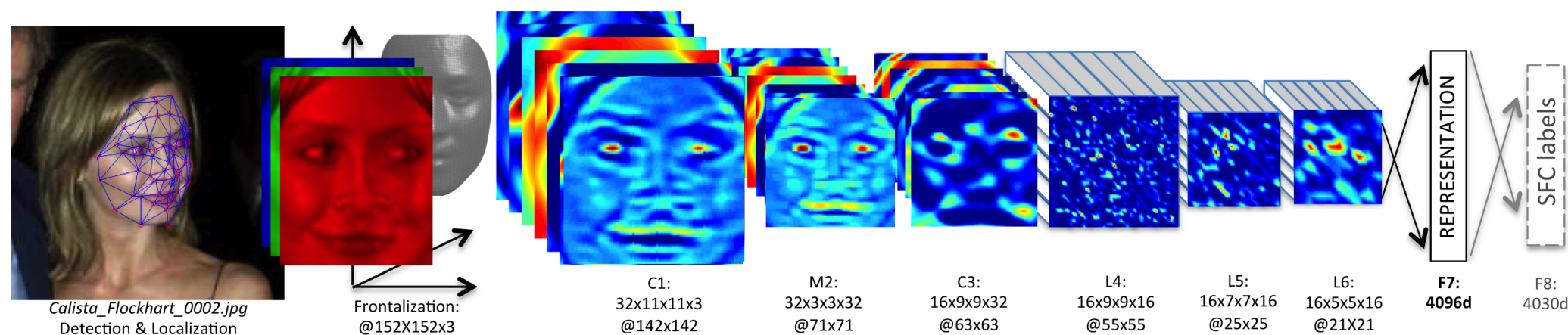
<https://www.youtube.com/watch?v=IMPjPQSb9g8&feature=youtu.be>

Some examples of successful DL applications ...

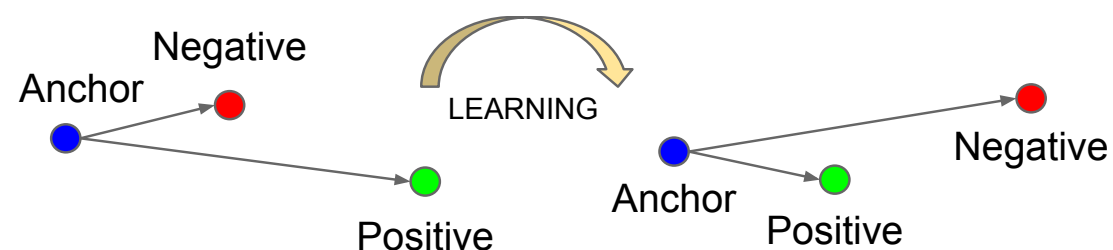
- **Face recognition**

People

- 4 million images of 4,000 persons



DeepFace: Closing the Gap to Human-Level Performance in Face Verification.
Yaniv Taigman Ming Yang Marc'Aurelio Ranzato Lior Wolf. CVPR 2014. Facebook



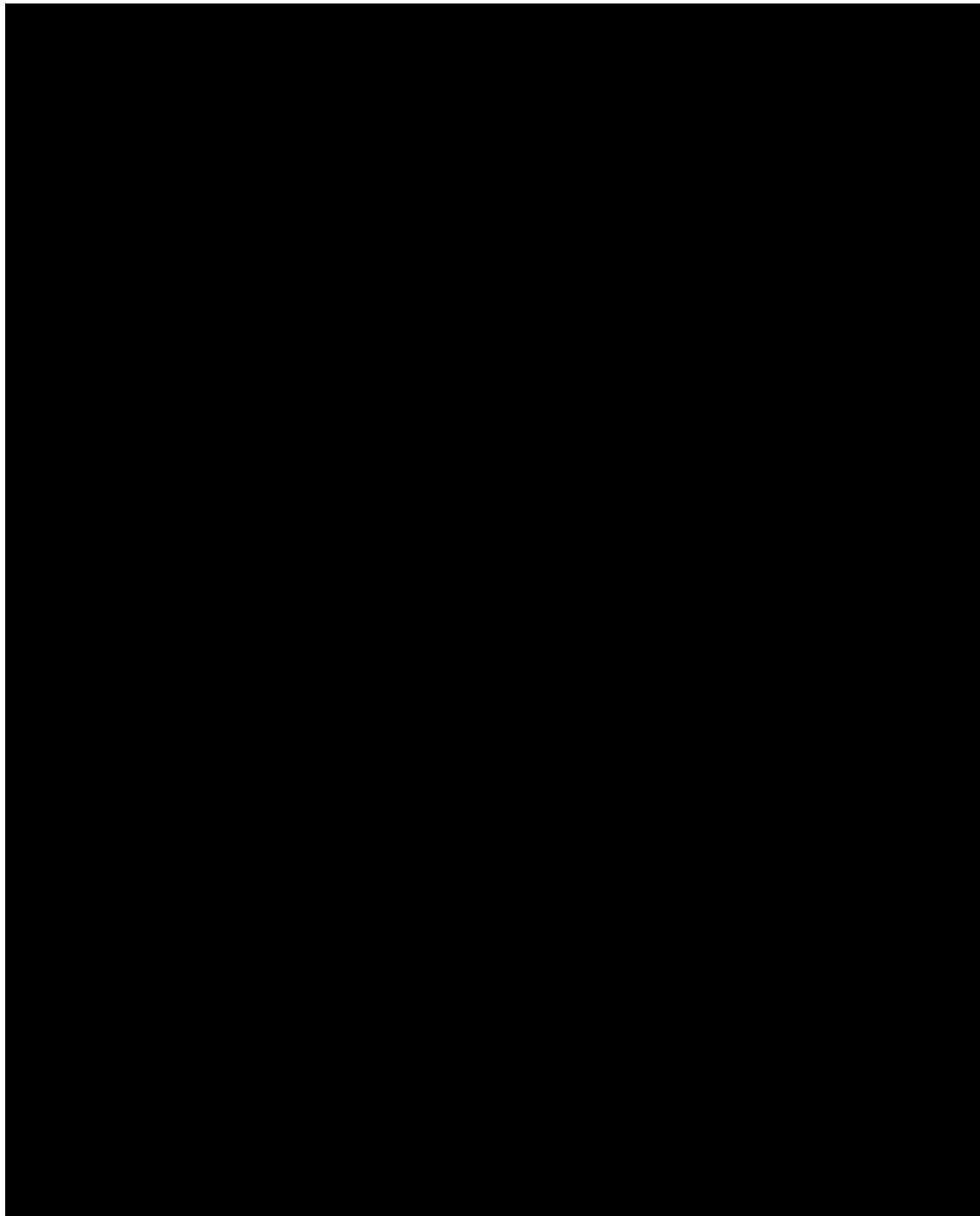
FaceNet: A Unified Embedding for Face Recognition and Clustering.
Florian Schroff, Dmitry Kalenichenko, James Philbin; CVPR 2015. Google



Some examples of successful DL applications ...

- Learn by watching how to play?! —> DRL

Behaviours

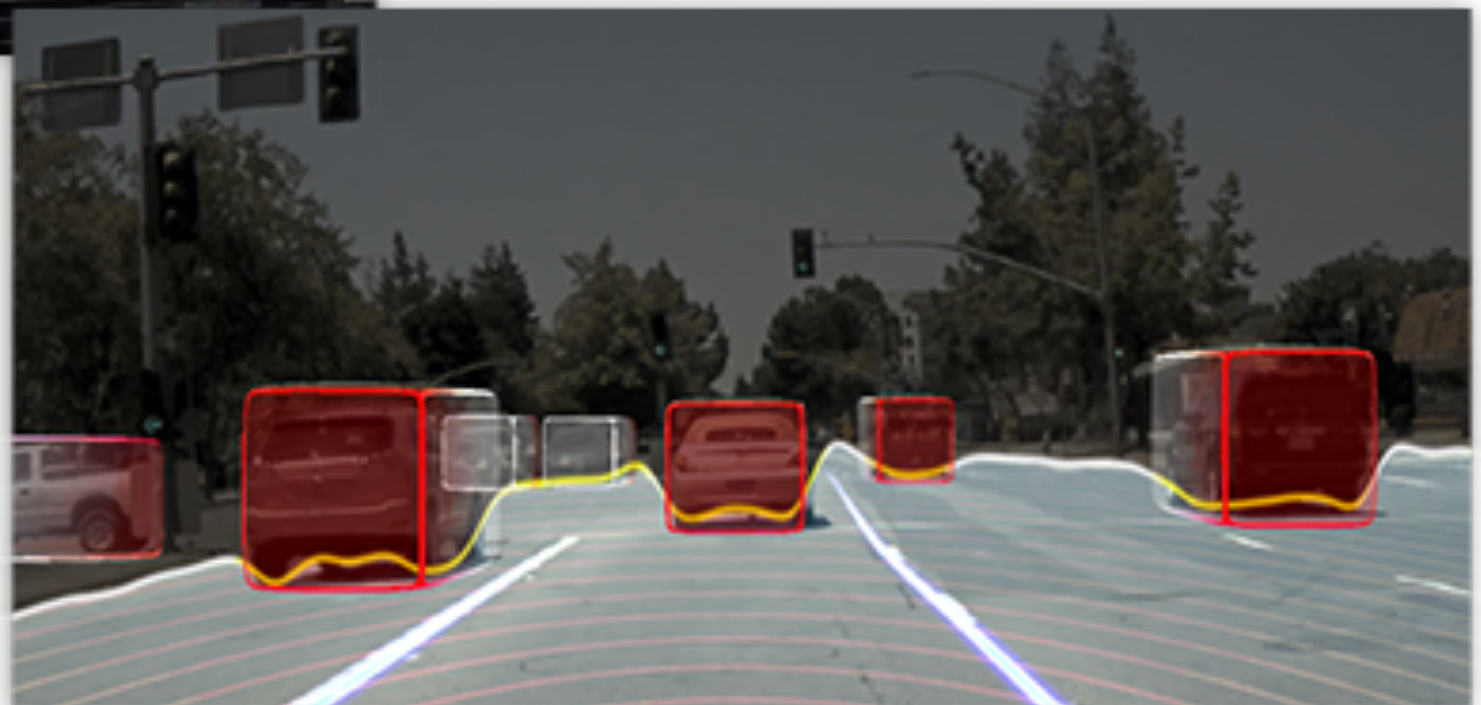


Google's Deep Mind

Some examples of successful DL applications ...

- Self-driving car

Behaviours



Healthcare

● Diagnosis

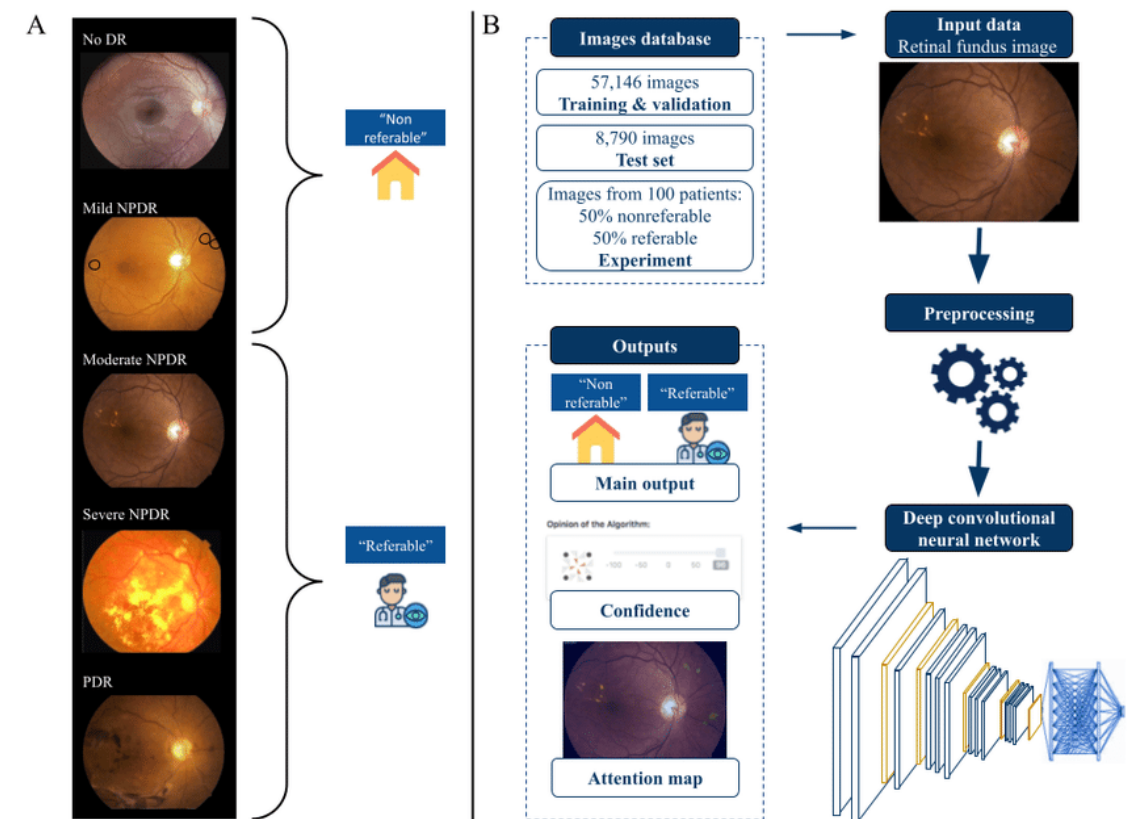
Ophthalmology (diabetic retinopathy): Deep CNNs on retinal fundus photos. Google's system (published in JAMA, 2016) matched specialists, and IDx-DR became the first FDA-authorized *autonomous* diagnostic AI in 2018.

Dermatology: Esteva et al. (Nature, 2017) trained a CNN to classify skin lesions at dermatologist level from photographs.

Radiology: CheXNet (a DenseNet) for pneumonia on chest X-rays; Google's mammography model (Nature, 2020) for breast cancer screening; Viz.ai's FDA-cleared stroke (large-vessel occlusion) detection on CT.

Pathology: CNNs on whole-slide histology for detecting lymph-node metastases (the CAMELYON challenge); Paige Prostate became an FDA-authorized cancer-detection tool. Newer work uses **vision transformers** and pathology **foundation models**.

Cardiology: ECG deep learning — Stanford's arrhythmia model and Mayo Clinic's ECG-AI, which infers things not visible to humans, like reduced ejection fraction or atrial fibrillation from a normal-rhythm ECG.



Healthcare

● Treatment development

Protein structure — AlphaFold (DeepMind): The landmark example. AlphaFold2 (2021) solved structure prediction using attention-based architecture (the Evoformer); AlphaFold3 (2024) extended this to protein–ligand and protein–nucleic-acid complexes. Related: RoseTTAFold and Meta's ESMFold (a protein language model).

Antibiotic discovery: Stokes et al. (Cell, 2020) used a **graph neural network** to screen millions of molecules and identified *halicin*, a structurally novel antibiotic.

Generative drug design: VAEs, GANs, and diffusion models for de novo molecule generation. Insilico Medicine used generative AI to design a fibrosis drug candidate that advanced into clinical trials.

Genomics: DeepVariant (Google) for variant calling, SpliceAI for splice-site prediction, and Enformer for predicting gene expression from DNA sequence.

AlphaFold structure prediction

The protein structure below has been predicted by [DeepMind](#) with AlphaFold ([Jumper, J et al. 2021](#)). For more information and additional features, please visit this sequence's page at [AlphaFold DB](#).

Information

Protein

P05067

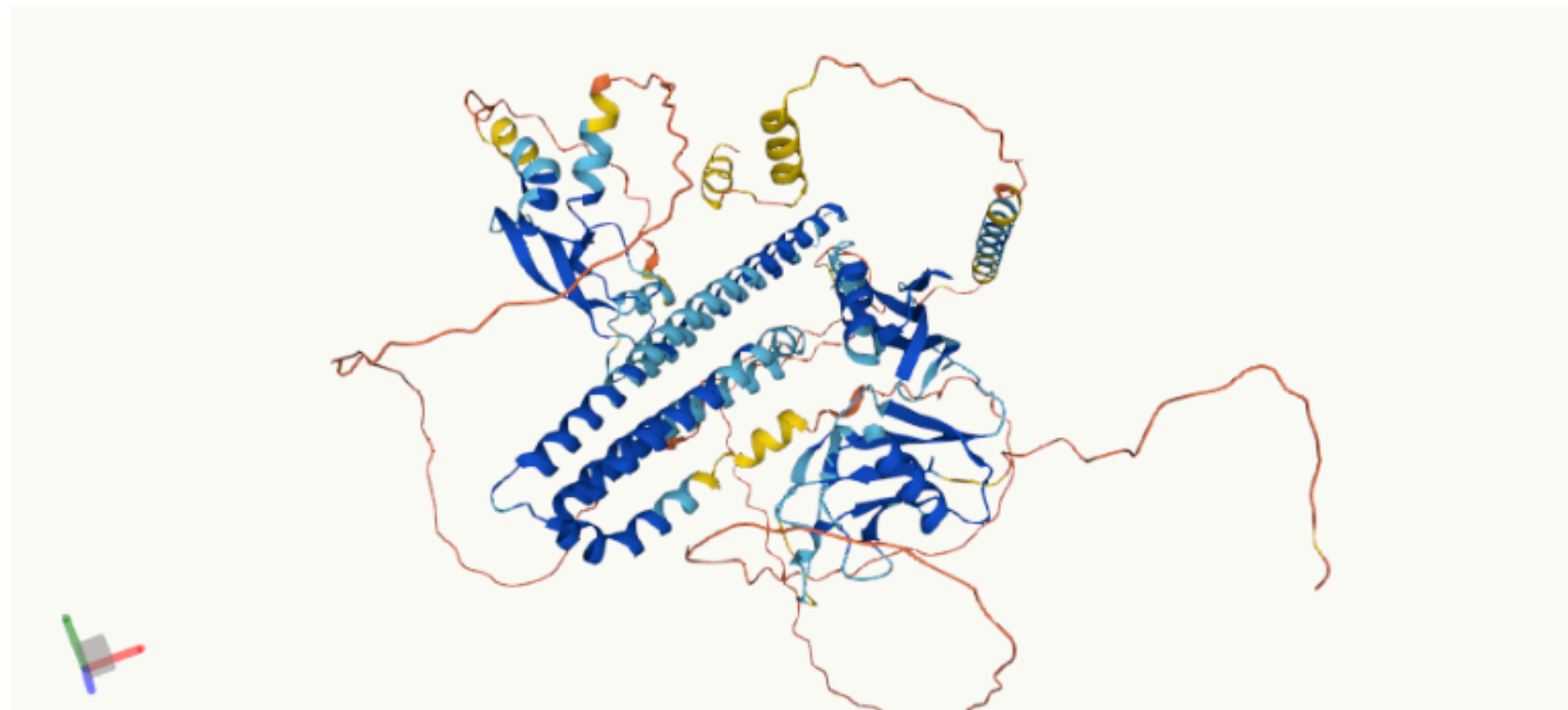
View on [AlphaFold DB](#) or [UniProtKB](#)

Organism

Homo sapiens

Model confidence

- Very High (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very Low (pLDDT < 50)



Healthcare

● Monitoring

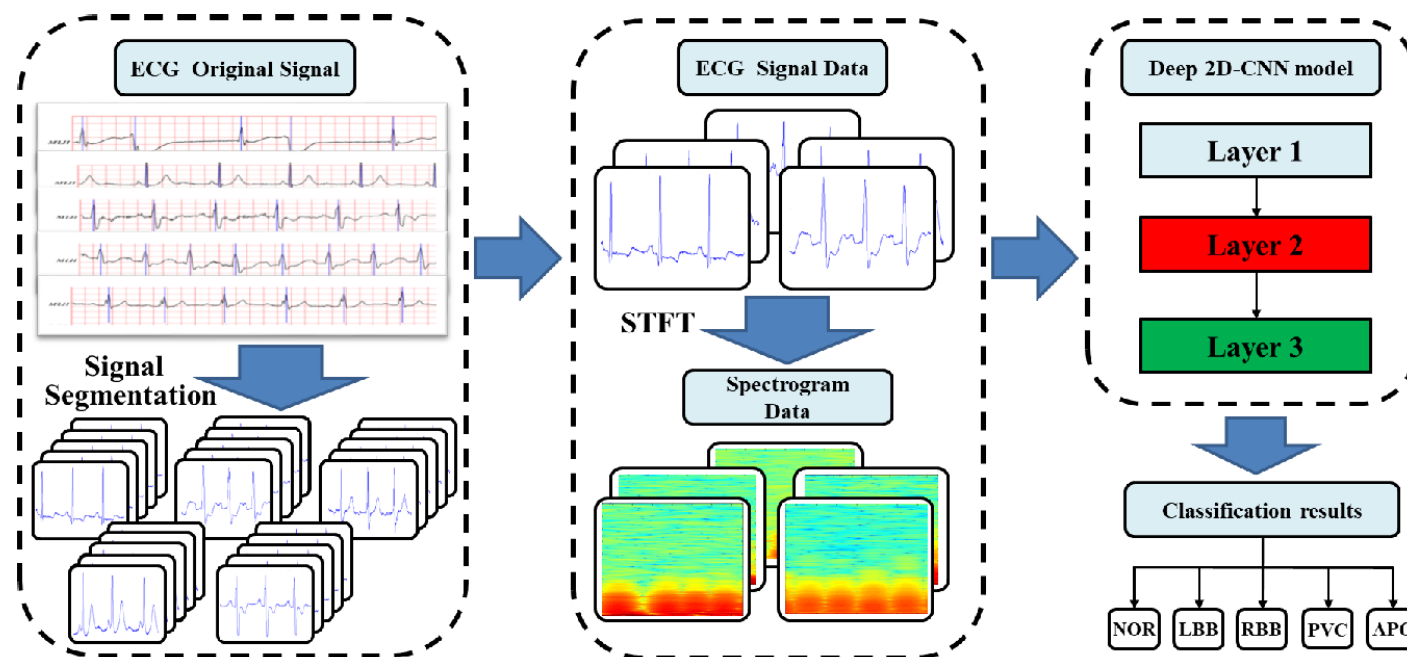
EHR-based risk prediction: Rajkomar et al. (npj Digital Medicine, 2018) applied deep learning to raw electronic health records to predict mortality, readmission, and length of stay.

Sepsis early warning: Duke's Sepsis Watch and Johns Hopkins' TREWS, the latter validated in a large real-world deployment (Nature Medicine, 2022) showing faster treatment.

Wearables: Apple Watch atrial-fibrillation detection uses deep learning on photoplethysmography (PPG) signals; similar approaches drive continuous glucose prediction and fall detection.

Critical care: Models predicting hypotension or clinical deterioration from ICU vitals streams.

Neurology: Deep learning on EEG for seizure detection and automated sleep staging from wearable signals.



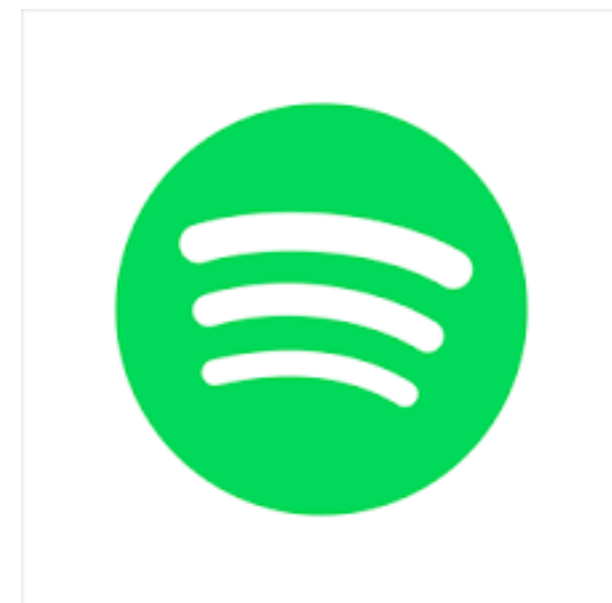
Some examples of successful DL applications ...

- DeepMind AI
- Reduces Google Data Centre Cooling Bill by 40%
- *“Using a system of neural networks trained on different operating scenarios and parameters within our data centres, we created a more efficient and adaptive framework to understand data centre dynamics and optimize efficiency.”*

Google's Deep Mind

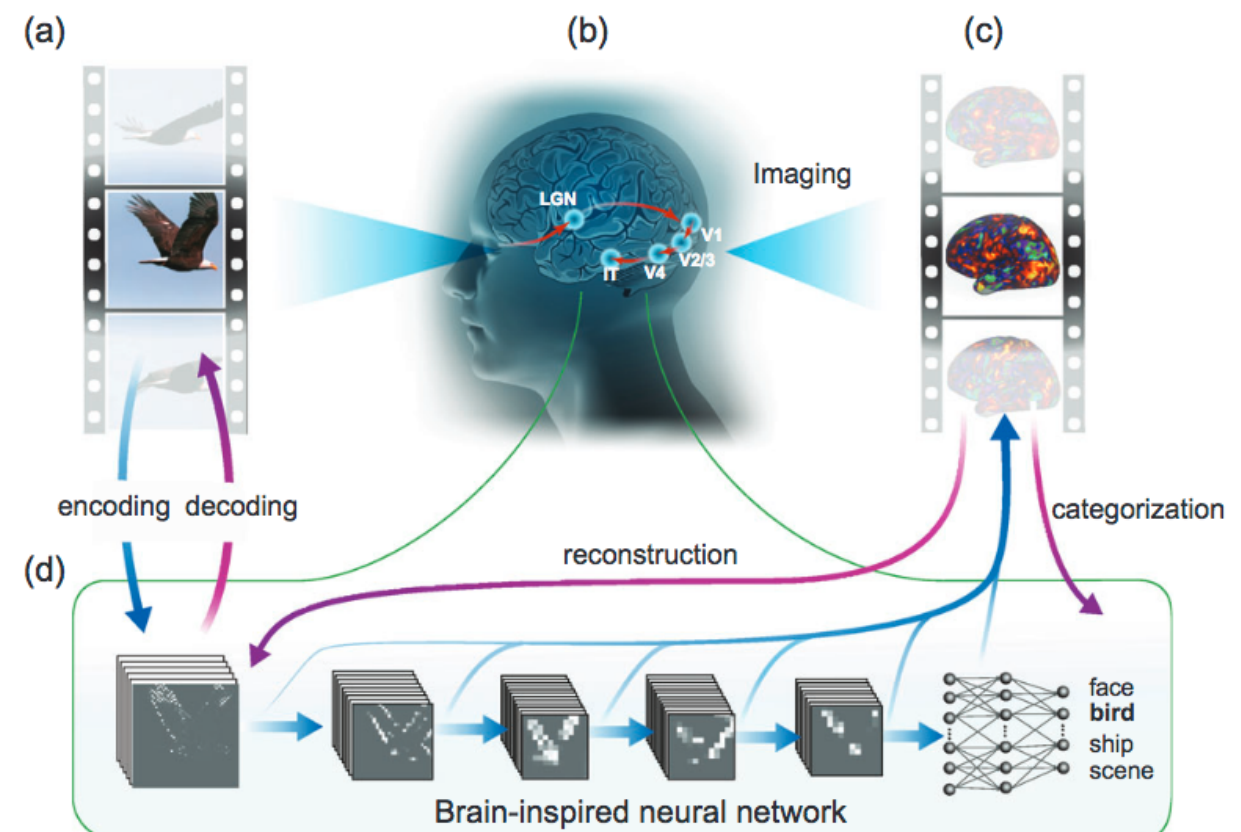
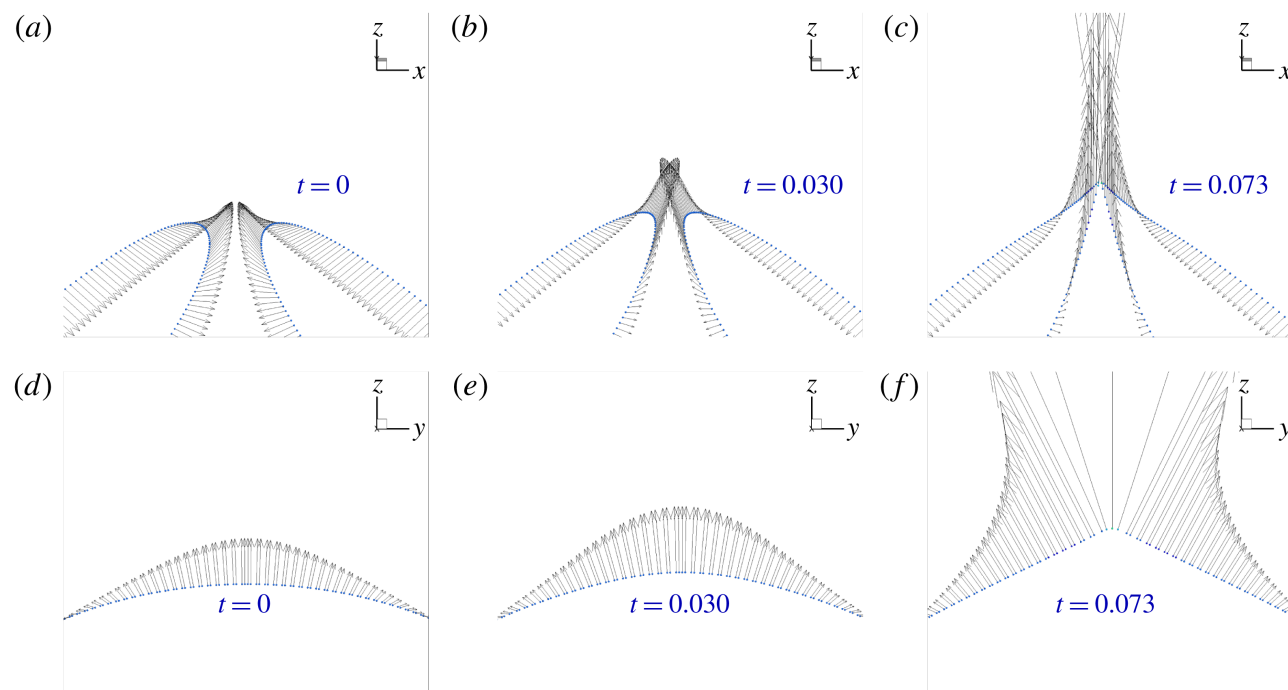
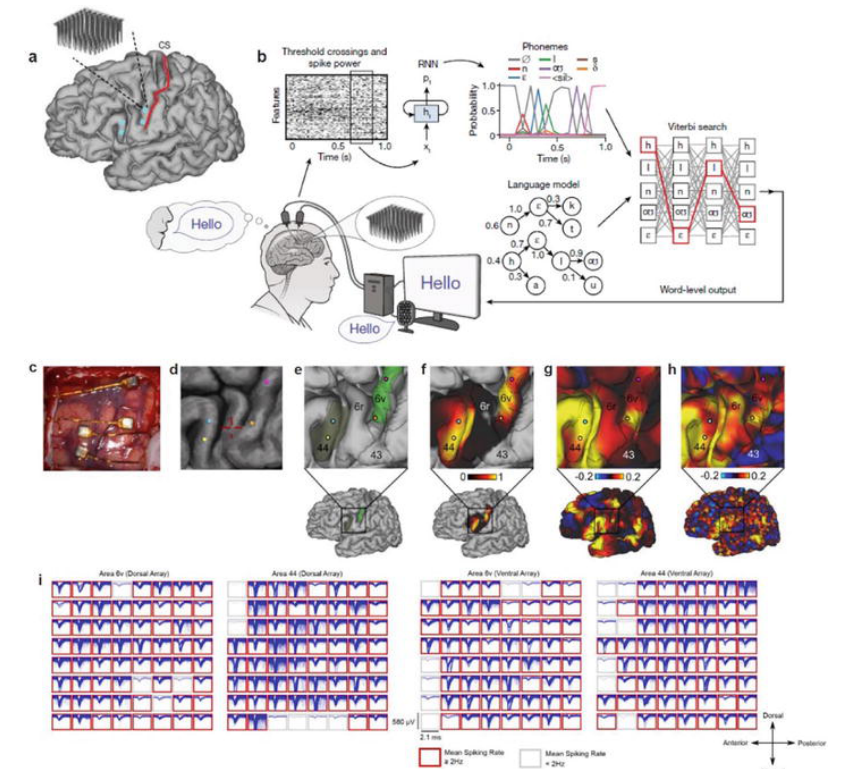
Some examples of successful DL applications ...

- **Recommendation** systems



Some examples of successful DL applications ...

- Science
- Math!
- Brain decoding!

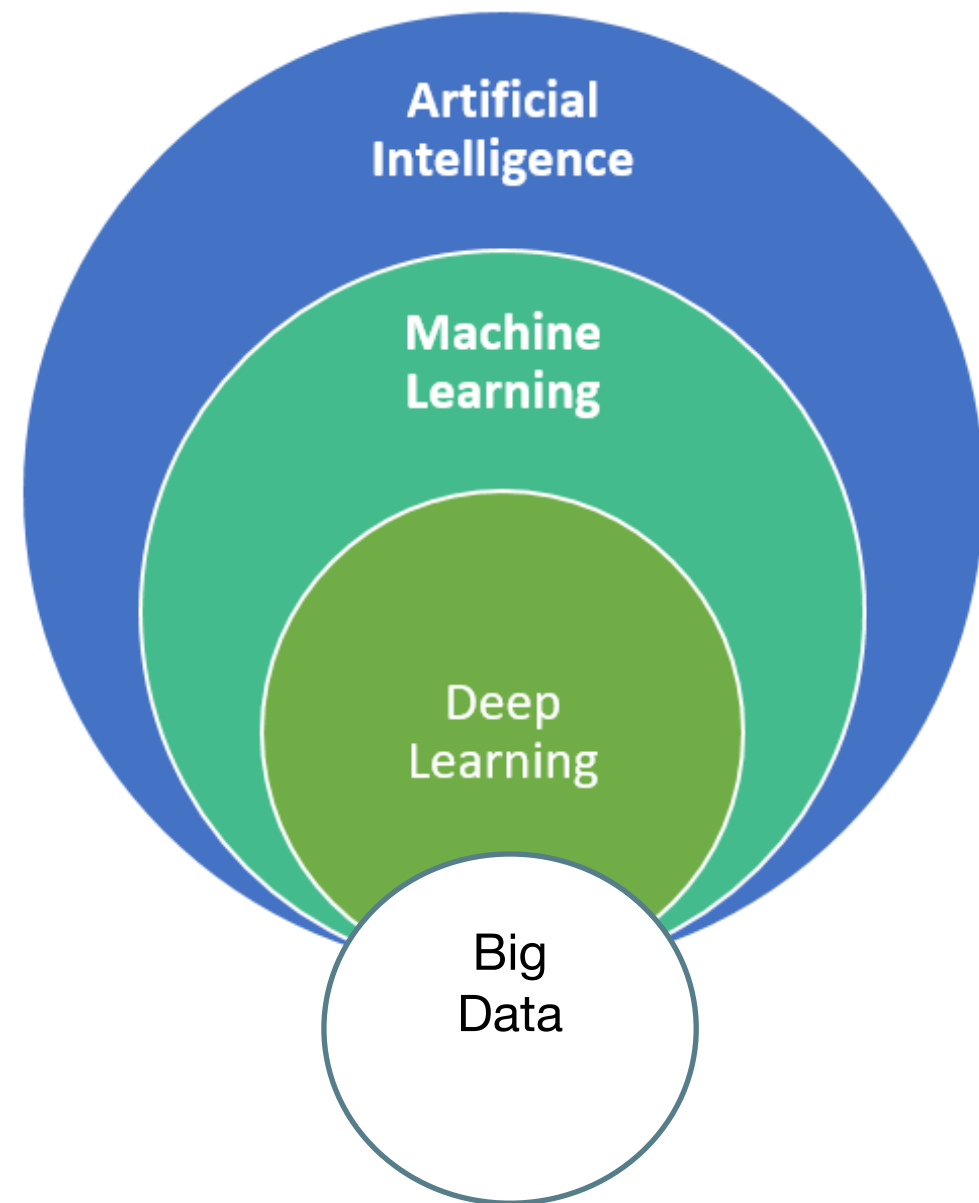


Some examples of successful DL applications ...

- Business products
 - **Coding**
 - **Marketing**
 - **Legal advice**
 - **Education**

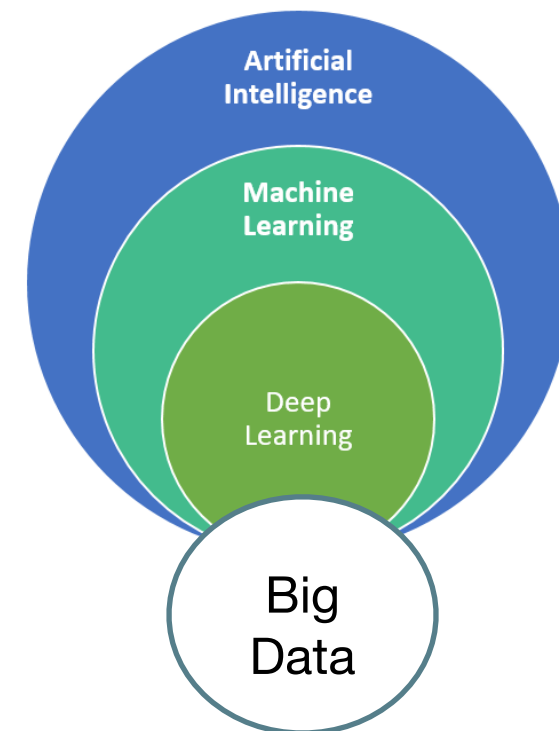
Deep Learning and Big Data

- Related topics to DL
 - Big data
 - Data science
 - Business Intelligence
 - ...



Deep Learning and Big Data

- Related topics to DL
 - **Big data:** *“data sets that are so large or complex that traditional data processing application software is inadequate to deal with them”* (Wikipedia)
 - **Data science:** Scientific field with the aim of extracting knowledge from raw data (usually “big data”)
 - **Business Intelligence:** Set of techniques to extract information from business data, that will be used to improve its performance.
 - *Data mining:* The exploration of raw data in the search for patterns (usually in the context of **databases**).

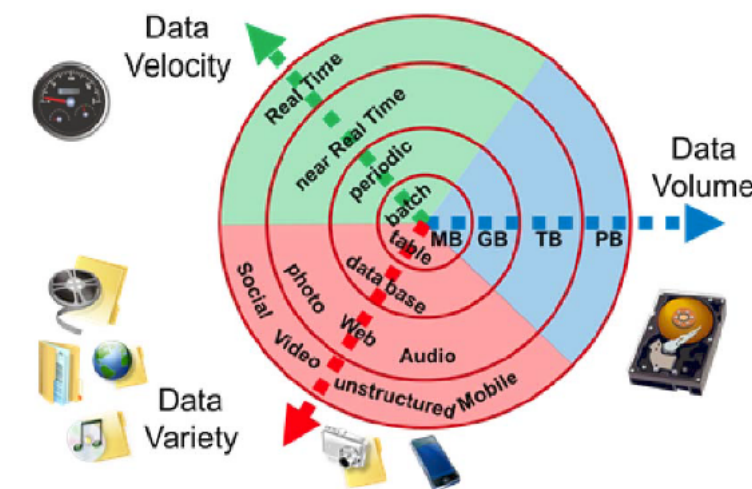


Data

- Dec'16 report from IBM Marketing Cloud (among others ...):
 - “90% of the data in the world created in the last two years”
 - “**2.5 quintillion bytes*** of data **a day**”
 - “Data growth rate will likely accelerate even more”
- more and **more** sources, **faster**, etc ...



- **By 2020, about 2.3 petabytes / second, 1% AI**
- **By 2026, about 4,7 petabytes (second, ???% AI**



- **Many companies in the data management business**

* (10¹² MB)

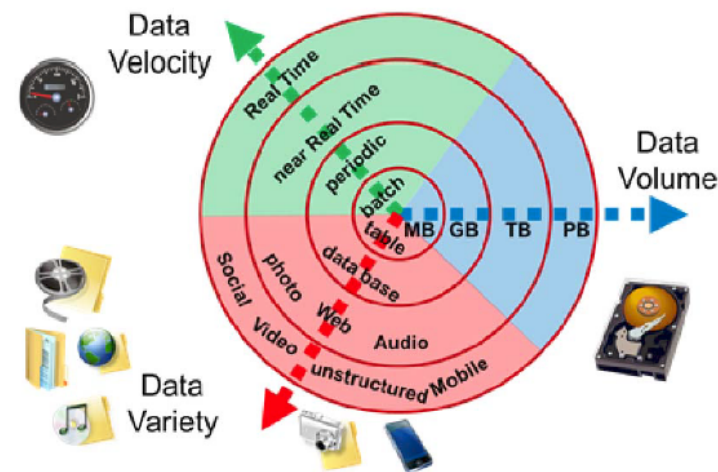
US million/billion/trillion/quart./quint. —> 10⁶, 10⁹, 10¹², 10¹⁵, **10¹⁸**

European million/billion/trillion/quart./quint. —> 10⁶, 10¹², 10¹⁸, 10²⁴, 10³⁰

Machine Learning + Big Data



+



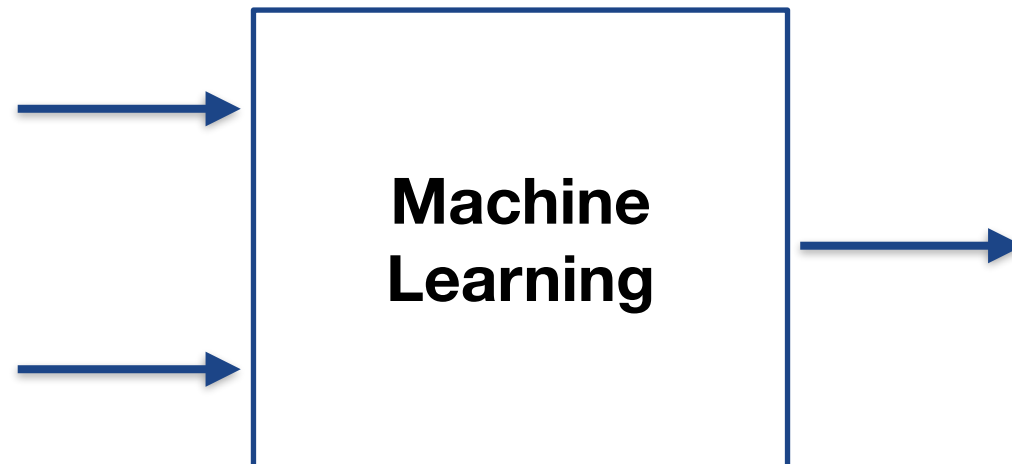
- **Cons:**

- Torture the numbers, and they'll confess to anything — Gregg Easterbrook (and probably many others ...)

- So it's like having billions of monkeys typing. One of them will write Shakespeare. — Michael Jordan

- **Pros:** New opportunities, challenges, and amazing results and applications!!

Deep Learning vs general ML pipeline



Deep Learning vs general ML pipeline

- Same methodology you saw in the previous sessions. Certain steps/ideas get particularly important: data selection, avoid overfitting, ...

1. Frame problem

2. Get data

3. Explore data

4. Prepare data

*It may sound you just
“throw” as much data as
you can ... but not just that!*

5. Explore promising models

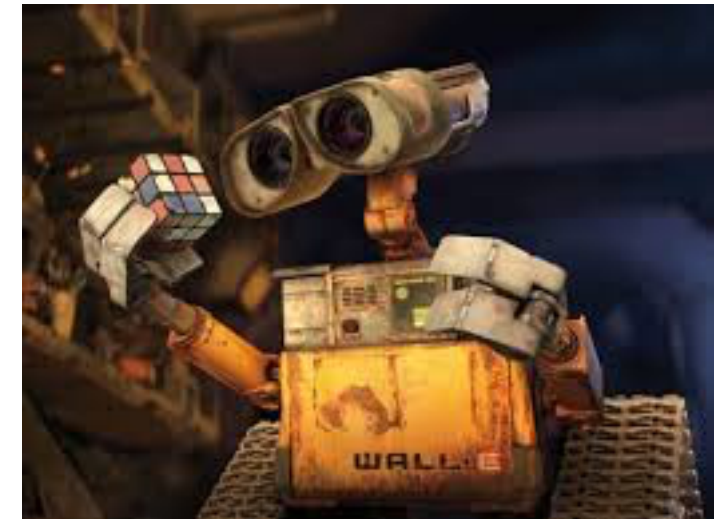
expensive!

6. Fine tune the best

7. Present/**analyze** the solution and results

ML pipeline & DL

- Get/explore/prepare data (labeled?), how to **represent** it?
- **Learn**: organize the sample data (training data): fit/learn a model
- **Classify: evaluate** new data with regard to the model learnt (classify/recognize new content)



How to get training data?

Lots of strategies. Crowdsourcing is a common ingredient lately

- ESP game. Luis Von Ahn and Laura Dabbish 2004
- LabelMe (MIT) - researchers and volunteers Russell, Torralba, Freeman, 2005
- Birds-200-2011 - fans of certain topic. Caltech-UCSD. Wah, Branson, Welinder, Perona, Belongie, 2011

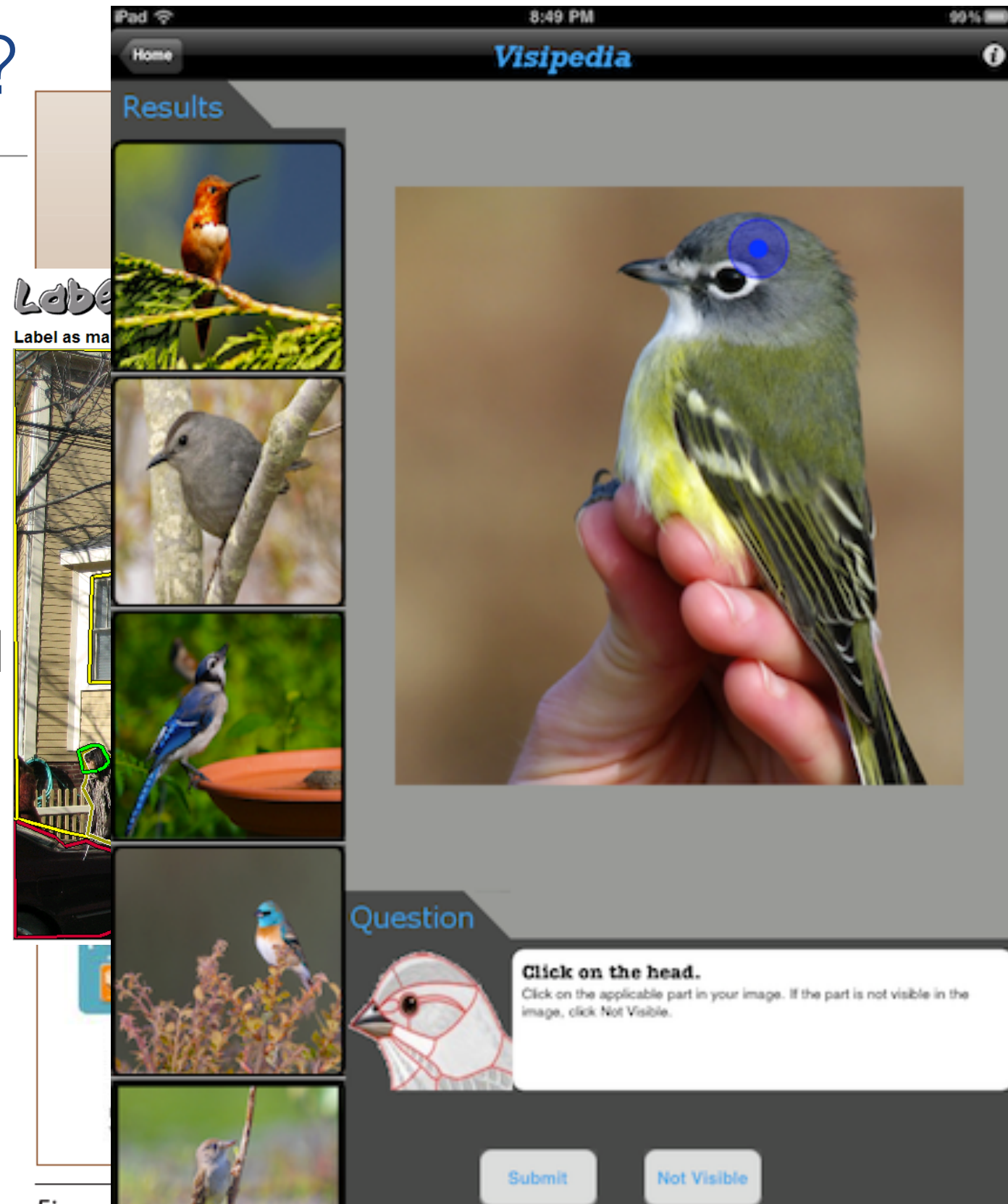
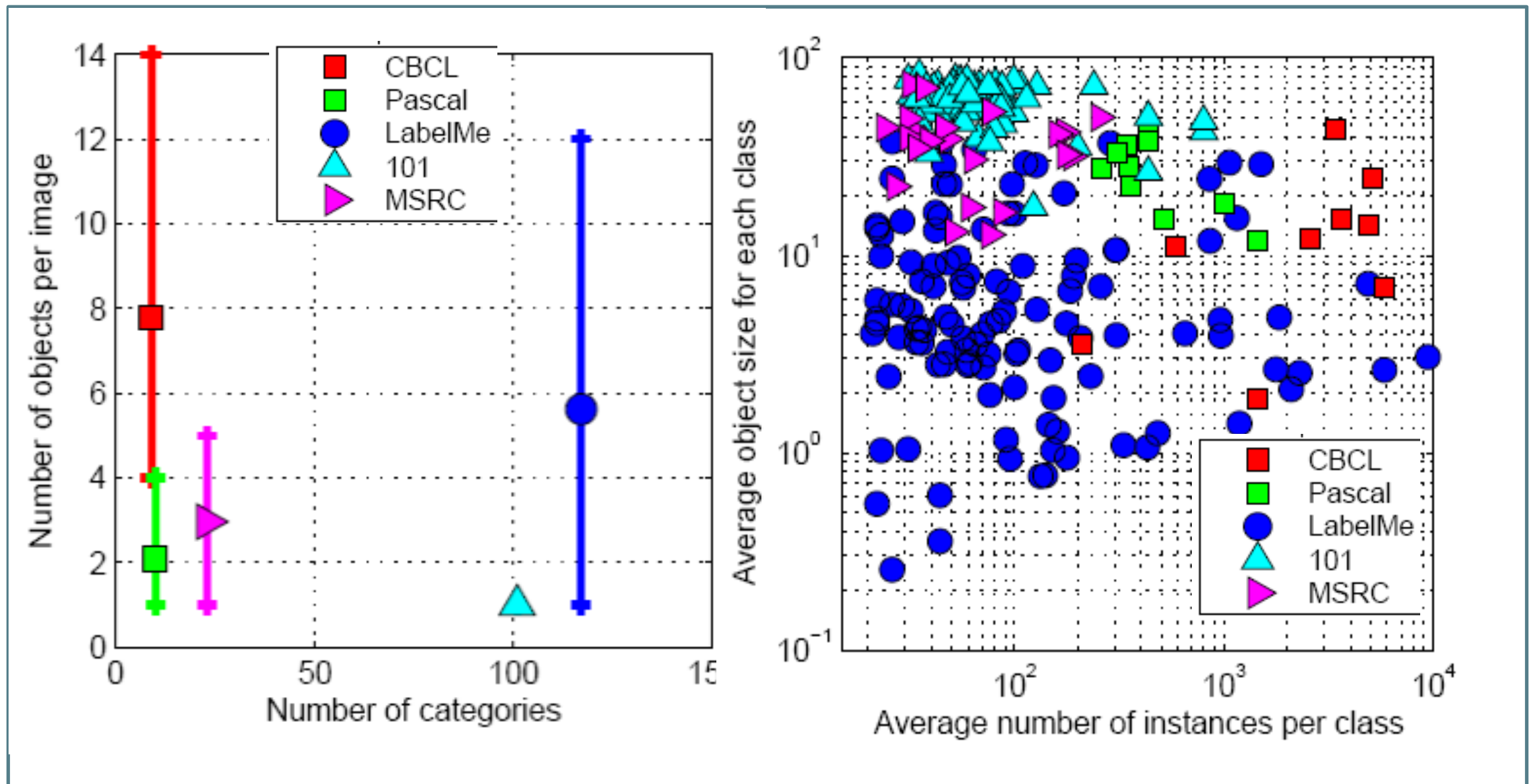


Figure 2. Peekaboom. Peek tries to guess the word associated with an image slowly revealed by "Boom."

Is my dataset good?

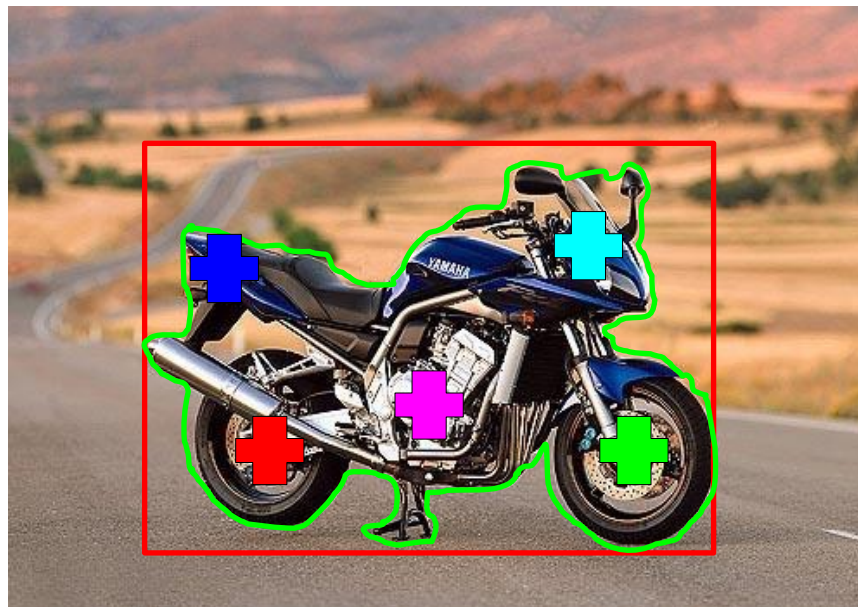
- Enough classes? Enough examples? Enough variability?
Good labels?



How to label training data?

- Depends on the application/problem

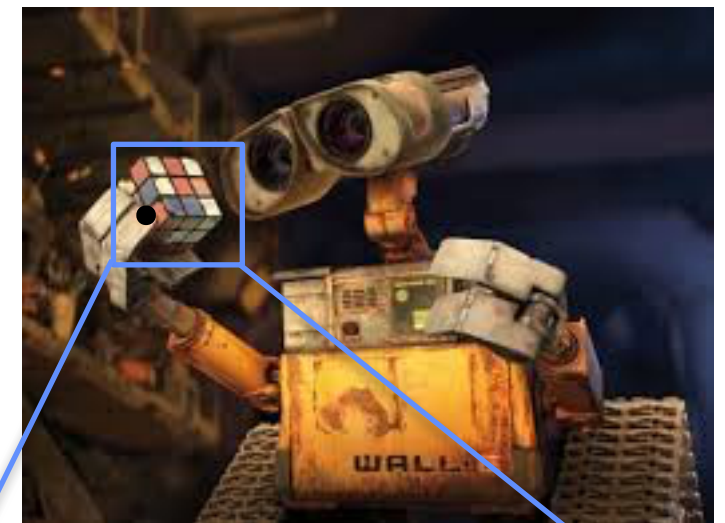
Contains a motorbike



ML pipeline & DL

- Get/explore/prepare **data** (labeled?), how to **represent** it?
- **Learn**: organize the sample data (training data): fit/learn a **model**
- **Classify**: **evaluate** new data with regard to the model learnt (classify/recognize new content)

features: VERY complicated and important *design* decision!!



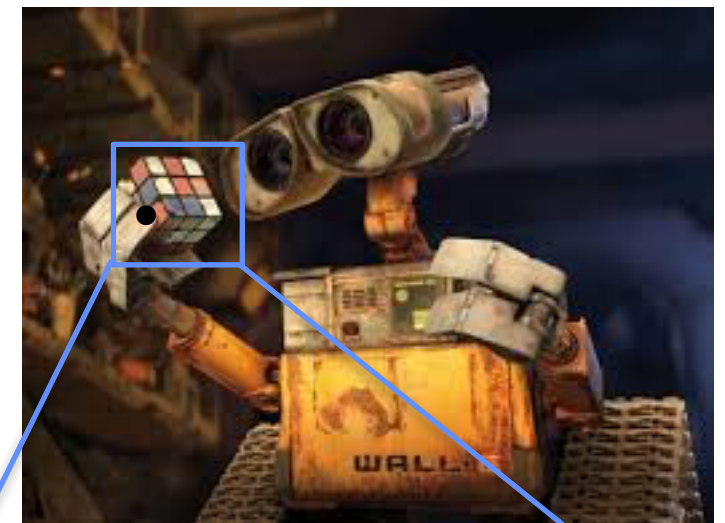
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ML pipeline & **DL**

- Get/explore/prepare data
(labeled?). ~~how to represent it?~~

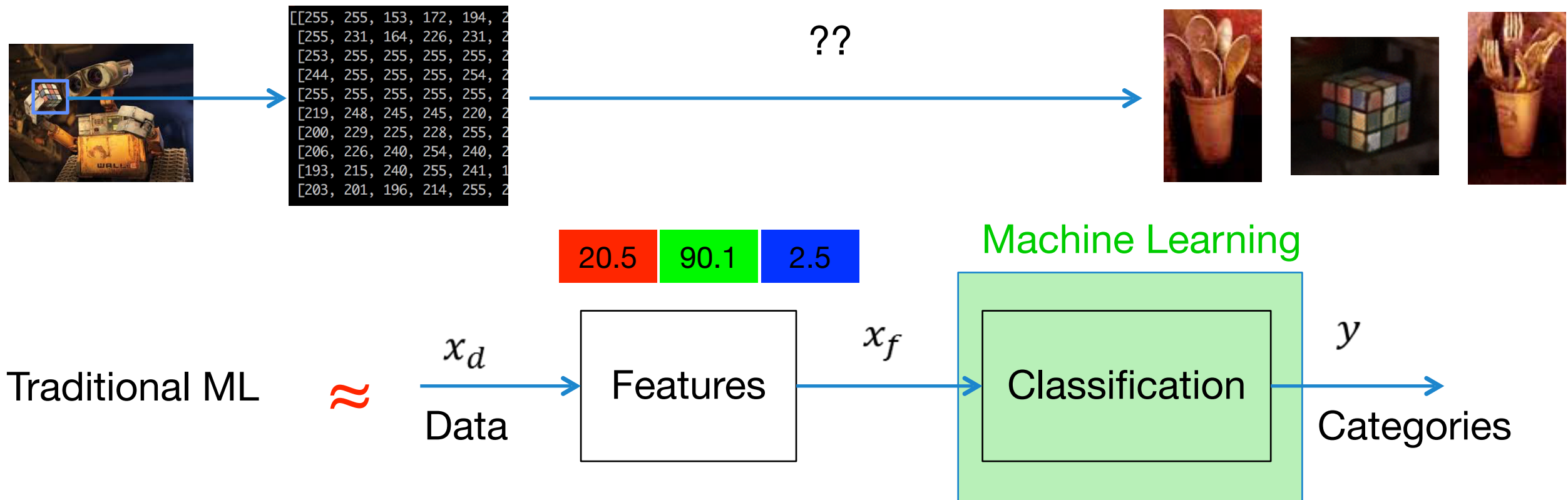
**features: VERY complicated
and important *design* decision!!**

- **Learn:** organize the sample data
(training data): fit/learn a model
- **Classify: evaluate** new data
with regard to the model learnt
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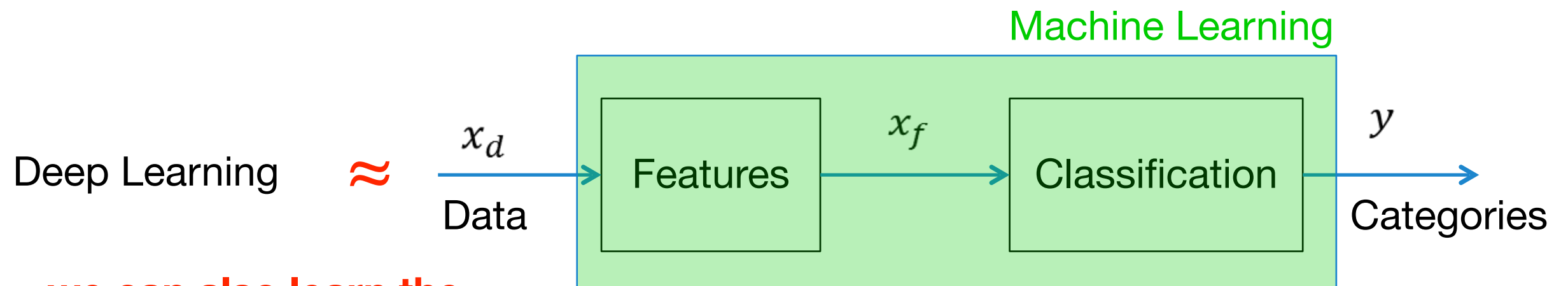
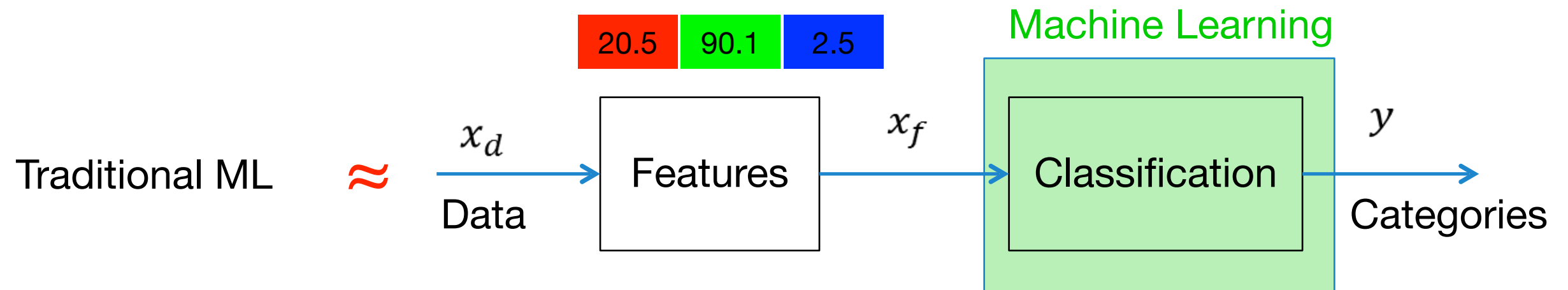


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```

ML pipeline & **DL** pipeline



ML pipeline & **DL** pipeline



we can also learn the representations (features)

Next

(Key “take-home” messages)

- What’s deep learning?
 - Related topics **(AI $\geq$ ML $\geq$ DL)**
 - DL pipeline **(Data? Able to learn suitable representation)**
- **Fundamentals of DL**
 - **Review basic concepts**
 - NN and DNN

Bibliography - Resources for some of the materials today

- Ian Goodfellow, Yoshua Bengio, Aaron Courville. ***Deep Learning***. MIT Press, 2016. <http://www.deeplearningbook.org>
- Book: ***Understanding Deep Learning***, Simon Prince, The MIT Press, 2023
- *Check many videos & tutorials*
 - *Stanford classes on deep learning for Computer Vision (<http://cs231n.stanford.edu>)*
 - *Ian Goodfellow, <https://www.youtube.com/embed//vi7IACCKOUao>*